



Centralized and decentralized autonomous dispatching strategy for dynamic autonomous taxi operation in hybrid request mode[☆]



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ABSTRACT

The combination of self-driving technology and taxis has a considerable potential in improving service quality and economic efficiency that can completely change traditional taxi operation. This study accordingly focuses on the autonomous taxi (aTaxi) dispatching problem in hybrid request mode where travelers can either request immediate rides or reserve taxi services ahead of time. In this study, a centralized dispatcher and decentralized autonomous dispatchers are designed to plan short-term and long-term routes for aTaxis in real time, respectively. The centralized dispatcher integrates vehicle-to-passenger assignment with empty vehicle rebalance to guarantee solution quality. Decentralized autonomous dispatchers distribute a portion of the calculation to aTaxis, thus reducing the centralized dispatcher's workload. The dispatching strategy ensures that the response to traveler requests can be made immediately, and if such requests are accepted, the travelers will receive the expected services. Finally, experiments are conducted based on the actual road network and the trip data of Manhattan to investigate the dispatching strategy. The results confirm the high performance of the proposed dispatching strategies in terms of service quality and economic efficiency. It is also found that the increase in reservation rates can enhance the robustness of the method with respect to errors in predicted requests. Moreover, when the number of vehicles is adequate and reservations are made longer ahead of time, the completion rate of requests and the revenue improve.

1. Introduction

In recent years, self-driving technology has rapidly developed. A large number of studies (Milakis et al., 2017) have confirmed the positive impact of the use of autonomous vehicles on road capacity, fuel efficiency, emissions, and accident risks. The combination of self-driving technology and taxis presents a promising prospect considering that aTaxis save labor costs and eliminate the constraint in the working hours of drivers. Moreover, aTaxis offer significant advantages in empty vehicle rebalance and operation in hybrid request mode where immediate requests and reservation requests coexist.

The spatial distribution characteristic of travel demand typically results in spatial distribution imbalance among vehicles that increases the waiting time of travelers in some areas. Although some current ride-hailing platforms, such as DiDi's heat map, can show the areas where numerous requests are generated so that drivers can be advised of their route, these would probably cause a large number of taxis to converge in certain areas. This because drivers generally follow the principle of maximizing self-profit. In contrast, idle aTaxis can be rebalanced by themselves in real time based on forecasted travel demands from the perspective of system

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optimum, thus improving service quality and system efficiency.

Moreover, aTaxi can potentially improve operational performance in hybrid request mode. At present, the taxi dispatching scheme of ride-hailing platforms lacks flexibility: once a reservation request is assigned to a taxi, it cannot be modified. However, with a dynamic variation of the system, the unmodified assignment cannot always be optimal. For example, before the scheduled departure time of an accepted reservation request, the vehicle may serve another immediate request, which requires travel to an area far from the origin of the previous reservation request. After completing the immediate request, the vehicle will have to travel over a long distance to return to the origin of the assigned reservation request. Actually, in this case, if the dispatcher could reassign another vehicle nearby to the reservation request, then unnecessary empty vehicle miles would be avoided. This method, however, is not suitable for human drivers, because repeated rerouting is not only confusing but also further makes it more difficult for them to execute the task. In contrast, aTaxi are highly controlled by the dispatcher; even the speed of each vehicle in the fleet can be centrally controlled (Wei et al., 2017). If the dispatcher reorganizes aTaxi routes and traveler assignments in real time, aTaxi can immediately respond and execute the task (Hyland and Mahmassani, 2018). This instantaneous reactive ability of aTaxi greatly increases the flexibility of dispatching in hybrid request mode. In addition, each aTaxi has computing and communicating capabilities that creates the possibility of achieving the proposed “decentralized autonomous dispatchers.”

This study focuses on the aTaxi dispatching problem with no ride-sharing. As far as we know, there is no previous research on the aTaxi dispatching problem in hybrid request mode. In this paper, the dispatching strategy in hybrid request mode is first proposed, which combines centralized dispatching and decentralized autonomous dispatching. Not only can this method achieve better service quality and economic benefits for service provider, but also computational efficiency are greatly improved. The contributions of this study are as follows.

- (1) An aTaxi dispatching strategy in single request mode where only immediate requests exist is proposed. The strategy can solve both the vehicle–passenger assignment problem and empty vehicle rebalance problem. It can also reroute aTaxi in real time with the dynamic variation of the system.
- (2) An aTaxi dispatching strategy in hybrid request mode is proposed. The dispatching strategy are realized by centralized dispatcher and decentralized autonomous dispatcher. Centralized dispatcher plans short-term routes for all aTaxi centrally, which enable aTaxi to travel in a better route. Decentralized autonomous dispatcher makes aTaxi manage its own long-term route, i.e., assigning newly arrived long-term requests and reserving capacity for accepted long-term requests. The dispatching strategy ensures that traveler requests can be responded immediately and get service if accepted.
- (3) Experiments based on actual road network and trips data of Manhattan are performed. The proposed dispatching strategies are tested in single request mode and hybrid request mode. The results confirm the better performance that the dispatching strategies can achieve and the great improvement in computational efficiency due to decentralized autonomous dispatchers.

The remainder of the paper is organized as follows. In [Section 2](#), existing literature related to aTaxi dispatching problem is reviewed. [Section 3](#) describes the aTaxi dispatching problem that this study focuses on in detail. [Section 4](#) presents the dispatching strategy in single request mode, whereas [Section 5](#) introduces that in hybrid request mode. [Section 6](#) discusses the experiments conducted to test the dispatching strategies. Finally, the conclusions are presented in [Section 7](#).

2. Literature review

2.1. Dynamic multi-vehicle pickup and delivery problem

According to Psaraftis et al. (2016), the aTaxi dispatching problem considered in this study can be classified as a dynamic multi-vehicle pickup and delivery problem with time constraints. A series of reviews on the dynamic vehicle routing problem, e.g., Berbeglia et al. (2010), Pillac et al. (2013), Ritzinger et al. (2015), Psaraftis et al. (2016), Molenbruch et al. (2017), Ho et al. (2018), has been performed.

The dynamic element that most studies consider pertains to travel requests that are revealed over time. To deal with dynamic characteristics, researchers propose two approaches. The first approach is to constantly resolve a static mathematical programming problem as new information arrives. This technique, which is similar to the rolling-horizon procedure, has been adopted by several studies reported in literature (Frantzeskakis and Powell, 1990, Fleischmann et al., 2004, Yang et al., 2004). This approach can achieve higher-quality solutions but consumes longer runtimes. The second approach is to update existing solutions by repairing heuristic approaches, such as insertion heuristics, while new information is revealed (Jaw et al., 1986, Calvo and Colorni, 2006, Luo and Schonfeld, 2007, Hame, 2011). This is utilized to achieve real-time computational efficiency; however, the solution derived through this technique is inferior to that of the first approach.

This study combines the advantages of both approaches. The first approach is employed to optimize the short-term route and enable vehicles to travel in a better route. The second approach is employed to plan the long-term route. When a request, whose departure time window is distant from the present period, is received, it is inserted into a planned route heuristically. Over time, the long-term route will become a short-term route; hence, it will be also re-optimized by the first approach, which guarantees the quality of the solution to be derived.

In view of the impact of the current solution on the fleet capacity to respond to future requests, some studies incorporate uncertain future conditions into current decision-making to achieve better solutions, as summarized in [Table 1](#). Godfrey and Powell (2002a), Godfrey and Powell (2002b), Novoa and Storer (2009) presented an approximate dynamic programming approach for dynamic fleet

Table 1
Comparisons between previous studies published in literature and this study.

Study	Request mode		Uncertain future conditions in objective function	Model		Algorithm
	Reservation request			Traveler-to-vehicle assignment	Empty vehicle rebalance	
	Immediate request					
Godfrey and Powell (2002a, 2002b)	✓	×	Approximate future value of resources	Integer program	–	Adaptive dynamic programming algorithm
Yang et al. (2004)	✓	×	Opportunity costs of serving jobs	Mixed-integer program	–	CPLEX
Zhang and Pavone (2015)	✓	×	Number of rebalancing vehicles on road	FCFS	Linear program	–
Zhang et al. (2016)	✓	×	Total rebalance time	Joint optimization, mixed integer linear program		Model predictive control algorithm
Chen et al. (2016)	✓	×	–	Dispatch nearest idle AV to traveler	Rule-based rebalancing strategies	–
Alonso-Moraa et al. (2017)	✓	×	–	Integer linear program	–	Maximum weighted matching algorithm
Ma et al. (2017)	×	✓	–	Linear program	–	CPLEX
Lowalekar et al. (2018)	✓	×	Expected value associated with future requests	Multi-stage stochastic optimization formulation introduced to advise vehicle rebalancing		Benders decomposition
Xu et al. (2018)	✓	×	Future state value of vehicle	Bipartite graph matching problem	–	Kuhn–Munkres algorithm
Hyland and Mahmassani (2018)	✓	×	–	Integer linear program	–	GUROBI
This study	✓	✓	Value of assigned predicted requests	Joint optimization, network flow model		CPLEX

management, which uses a nonlinear function to approximate the future value of resources. Yang et al. (2004) considered future requests in terms of the opportunity cost to complete accepted jobs. Bent and Van Hentenryck (2004) and Lowalekar et al. (2018) considered multiple scenarios, including known and future requests in route planning. Ghiani et al. (2012) compared the sample scenario planning approach and online anticipatory algorithm for dynamic and stochastic routing problems in terms of solution quality and computational effort. Xu et al. (2018) assigned requests to drivers in real time to maximize the sum of instant reward and future-state value, where future-state value is estimated by the temporal difference based on historical data.

Compared with studies that utilize the probability distribution of future request generation as model input, this research incorporates explicit predicted requests with origin, destination, request time period, and value. There are three reasons for this incorporation. First, the considerable progress achieved in working with big data and predictive analytics that diminishes uncertainty in future information makes obtaining explicit predicted requests possible. Second, the explicit predicted request can also represent the predicted request from one zone to another with the request time within a time period, consequently reducing prediction difficulty. Third, in the proposed model, it is more suitable to plan explicit predicted requests simultaneously with immediate requests and reservation requests.

2.2. Taxi dispatching problem and taxi route recommendation problem

The taxi dispatching and taxi route recommendation problems are research areas in the dynamic multi-vehicle pickup and delivery problem, which are considerably related to the problem that this study focuses on. The research on these two problems is reviewed as follows.

In terms of the taxi dispatching problem, a number of works concentrate on reducing the waiting times of travelers. Lee et al. (2004) assigned taxis to travelers with shortest travel times based on real-time traffic conditions. Maciejewski et al. (2016a) applied the formulation of the assignment problem to taxi dispatching and compared it with the demand–supply balancing strategy. Maciejewski et al. (2016b) proposed a demand–supply balancing strategy where requests are served immediately by the nearest taxis with low demand, and vehicles are dispatched to the nearest requests that await service once they become idle under high demand conditions. Zhu and Prabhakar (2017) designed a trip assignment model based on network flow. Apart from these studies, there are some works that focus on other performances. Ngo et al. (2004) formulated the taxi assignment problem as a fuzzy linear assignment problem, where the cost value of a request “fuzzily aggregates” multiple criteria defined by human experts. Gao et al. (2016) designed a new mobile taxi-hailing system, which incorporates the total net profits of taxis and waiting time of travelers into a system utility function. Situ et al. (2017), Liu et al. (2017) proposed a novel region-dependent decomposition strategy to deal with the large-scale taxi dispatching problem and divided the service area into several sub-regions, each of which can match travelers for taxis in parallel. Ramezani and Nourinejad (2018) designed a holistic taxi dispatching system with the interaction between taxis and normal traffic considered, where the macroscopic fundamental diagram approach is used to represent the dynamic evolution of traffic conditions and the model predictive control approach is used to solve the optimal taxi dispatch control problem. Nourinejad and Ramezani (2019) investigated a dynamic non-equilibrium ride-sourcing system where drivers may join or exit the market based on their perception of wage and search time, and proposed pricing strategies to maximize the service provider’s profit.

Some researchers incorporate human characteristic into modeling the taxi dispatching problem. To bridge the revenue gap among taxi drivers and ensure a short waiting time for travelers, Dai et al. (2017) focused on fair assignment and proposed a balanced assignment mechanism for online taxi recommendation. Additionally, a few human-driven taxi dispatching strategies proposed by researchers allow taxi reassignment. Glaschenko et al. (2009) presented a taxi re-scheduling method to achieve better system performance. In the reassignment, taxi drivers have to negotiate among themselves. With respect to changes in the expected incomes of drivers that result from reassignment, Billhardt et al. (2019) designed an economic compensation scheme to persuade taxi drivers to accept reassignment that may lead to a better solution from a global perspective.

A number of studies focus on human-driven taxi route recommendation similar to the empty vehicle rebalance problem for autonomous vehicles. The taxi route recommendation aims to aid drivers in planning their cruising routes. This not only enables drivers to pick up more passengers but also reduces the waiting time of passengers. According to Lai et al. (2019), the taxi route recommendation is divided into two classes: macroscopic and microscopic recommendations. The macroscopic recommendation only provides driving directions to taxi drivers, such as those reported by Powell et al. (2011), Qian et al. (2012), Li et al. (2012), and Hsueh et al. (2014). They predicted profitable locations to guide taxi cruising, which may reduce taxi cruising time and increase revenues. Most macroscopic recommendations, however, ignore the competition among vacant taxis. The microscopic recommendation provides taxi drivers with detailed driving routes. Zhang and He (2012) designed a cruising system, pCruise, to recommend the shortest cruising route with at least one passenger expecting to take a taxi. Dong et al. (2014) used the real-world global-positioning system data to evaluate the score of each road segment and calculate profitable cruising route based on them. Qian et al. (2015) focused on recommendation fairness among competing taxi drivers and proposed a sharing-considered route assignment mechanism for fair taxi route recommendations. Liu et al. (2015) presented non-myopic adaptive routing algorithms to minimize the collective travel time of all vehicles by considering uncertain and dynamic congestion conditions. Luo et al. (2018) considered the dynamic change in capacity and probability along the recommended route after a taxi responds to a request. They also designed an algorithm to plan cruising routes and update them in real time. Lai et al. (2019) modeled the relationship between taxis and passengers with Coulomb’s law, the attractiveness between taxis and passengers, and the competition among taxis, from which more accurate and effective route recommendations can be derived.

Compared with human-driven taxis, the aTaxi dispatching problem is easier to resolve because it does not consider human characteristics. On the one hand, when dispatching aTaxi, it is unnecessary to take into account the revenue fairness and competition

relationship among vehicles. On the other hand, it is not necessary to incorporate the system uncertainty resulting from driver behavior into the aTaxi dispatching. In human-driven taxis, the drivers can enter and leave the system at any time. They may disobey the dispatching strategy or route recommendation that may induce system uncertainty. In view of this, it is necessary for human-driven taxis to persuade some drivers to accept the dispatching strategy that may require them to sacrifice self-benefit but can aid them achieve the global optimal by certain economic methods. However, such a problem does not exist in autonomous taxis, because of high control. Additionally, without the constraint of human characteristics, autonomous taxis coordinate with each other and can achieve better performance from a global perspective.

2.3. Shared autonomous vehicle fleet operation

In recent years, investigations that pertain to shared autonomous vehicle (AV) fleet operation have been implemented. Hyland and Mahmassani (2017) presented a new taxonomy for shared autonomous vehicle fleet management problems based on existing taxonomic categories on vehicle routing and scheduling problems. The two common problems that researchers focus on are vehicle-to-passenger assignment problem and empty vehicle rebalance problem.

In the vehicle-to-passenger assignment, several researchers adopt heuristic rules. For example, Fagnant and Kockelman (2014), Fagnant et al. (2015), Chen et al. (2016), Bischoff and Maciejewski (2016), Jäger et al. (2017) dispatched the nearest idle AV to travelers. Levin et al. (2017) preferred to assign en-route vehicles to travelers to avoid sending more AVs. Pavone et al. (2012), Zhang and Pavone (2015), Figueira et al. (2016) assumed that vehicle assignment follows the first-come first-served (FCFS) rule at a station, where the passenger with the longest waiting time is the first to be served. Liu et al. (2019b) focused on finding reliable path for shared autonomous taxi to improve the probability of on-time arrival. In Liu et al. (2019b), newly arrived customer is matched to the feasible aTaxi with the maximal on-time arrival reliability based on the rules they defined.

Other researchers used an optimization model to find a system optimal match between the vehicle and traveler. Alonso-Moraa et al. (2017) built a general mathematical model for dynamic trip-vehicle assignment with ride-sharing. Vazifteh et al. (2018) presented a batch model to solve the vehicle dispatching problem online and used a maximum matching algorithm to find a current optimal solution. Hyland and Mahmassani (2018) allowed the reassignment of en-route pickup AVs and incorporated en-route drop-off AVs in the assignment. They found that their AV-traveler assignment strategies can reduce fleet miles and traveler waiting times compared to simple assignment strategies obtained by experiments. Liu et al. (2019a) integrated the vehicle routing problem and dynamic traffic assignment as space-time-state network flow models. They used the Dantzig-Wolfe decomposition and Column pool-based approximation to improve computational efficiency. Simonetto et al. (2019) proposed a more efficient ridesharing algorithm via linear assignment problems based on the study (Alonso-Moraa et al., 2017), which can be distributed into multiple ridesharing companies.

In addition, considerably few researchers focus on autonomous vehicle-sharing and reservation systems where travelers request AV trips ahead of time. Ma et al. (2017) proposed a linear programming model to plan the pickup and delivery schedules for AVs based on requests accepted in advance that can be solved efficiently.

In terms of empty vehicle rebalance, some researchers adopt rule-based rebalancing strategies. Fagnant and Kockelman (2014) divided the city into blocks, each of which is assigned an attribute “block balance” that reflects the difference between vehicle supply and demand in the succeeding 5-min period. The block with a high positive block balance value will push vehicles to adjacent blocks, and that block with low negative block balance value will pull similarly free vehicles from adjacent blocks until all block balances are controlled in the threshold. Fagnant et al. (2015), Chen et al. (2016) adopted the same vehicle relocation strategies as Fagnant and Kockelman (2014).

The heuristic rules may result in a suboptimal system performance. Some studies published in literature adopted an optimization model to solve the empty vehicle rebalance problem. Several researchers investigate an autonomous mobility-on-demand (AMOD) system where travelers have to arrive at a station to obtain service. For every station to reach equilibrium, Pavone et al. (2012) presented a fluid model, and Zhang and Pavone (2015) casted the AMOD system into the Jackson model. They proposed a rebalancing algorithm from a queueing-theoretical perspective. Figueira et al. (2016) tested different rebalancing strategies and confirmed the benefit of rebalancing. Pavone (2016) summarized the control approaches of the AMOD system, including the lumped approach and distributed approach. Chen et al. (2017) designed a hierarchical framework where higher hierarchies are responsible for vehicle rebalancing, and lower hierarchy plans the pick-up and drop-off schedules for each vehicle.

The foregoing studies published in literature use the remaining empty vehicles for rebalancing after a vehicle-to-passenger assignment. Only a few researchers adopt the joint optimization of vehicle-to-passenger assignment and empty vehicle rebalance, allowing the rejection of known requests to reserve the capacity for future higher-revenue requests. Zhang et al. (2016) proposed a model predictive control approach, whose decision variables are vehicle behaviors. It allows the vehicle to select rebalancing to another station or waiting at a station instead of serving a customer. This study also adopts joint optimization by assigning predicted and known requests simultaneously. When a predicted higher-revenue request occurs, the dispatcher may assign an empty vehicle to it even if there is an existing unserved request nearby. In addition, because of the uncertainty of a predicted request, a parameter is set to discount the value of the predicted request to reduce the number of rejections of existing requests.

The above studies considerably contribute to the shared autonomous vehicle fleet operation. However, there is no previous research that deals with the dynamic vehicle routing problem in hybrid request mode because of the difficulty of simultaneously planning immediate requests and reservation requests, as summarized in Table 1. On the one hand, reservation requests are expected to be responded to immediately, but the standard is vague for assessing how good the reservation request assignment is because of the uncertainty of the future state and location of the vehicle. On the other hand, the available time period of a vehicle is divided into

Table 2
Notations.

Notations	Definition
Sets	
V	Set of aTaxi that are available within short-term horizon
R	Set of short-term requests
R_I	Set of immediate requests
R_P	Set of short-term predicted requests
R_{Rn1}	Set of newly arrived short-term reservation requests
R_{Ra1}	Set of previously accepted short-term reservation requests
R_{Rn2}	Set of newly arrived long-term reservation requests
R_{Ra2}	Set of previously accepted long-term reservation requests
Request information	
i	Index of request or request node in aTaxi-request network
o_i	Origin of request i
d_i	Destination of request i
te_i	Earliest departure time of request i
tl_i	Latest departure time of request i
f_i	Profit that request i provides to the operator
a	Believability of prediction
ATaxi information	
k	Index of aTaxi or aTaxi node in aTaxi-request network
ta_k	Available time of aTaxi k
s_k	Available location of aTaxi k
ATaxi-request network	
P_{ij}	Profit on arc from node i to node j
x_{ij}	Flow on arc from node i to node j
Simulation parameters	
Δt	Updated interval of centralized dispatcher
T_{hor}	Length of short-term horizon
t	Current time
$t(l_1, l_2)$	Travel time from location l_1 to location l_2
$c(l_1, l_2)$	Travel cost from location l_1 to location l_2

multiple segments by the estimated occupied time of accepted reservation requests, thus complicating route planning. The emergence of autonomous vehicles provides a good opportunity to simplify these two problems. By taking considerable advantage of the instantaneous reactive ability of autonomous vehicles, an aTaxi dispatching strategy in hybrid request mode, where reservation requests may be made at any time before the departure time, is formulated.

3. Problem statement

The notations used in this paper are listed in Table 2.

Our focus is set on an aTaxi system with no ride sharing, and the aTaxis are functionally homogenous. In this system, the aTaxis are operated in a free-floating mode without stations, and vehicles are allowed to park anywhere in the service area. As illustrated on the left of Fig. 1, the dispatcher constantly collects travel requests (including immediate requests, reservation requests, and predicted requests based on historical data) and acquires information regarding the aTaxi status. The problem is how to route aTaxis (sequence of pickup locations, drop-off locations, and rebalance locations) in real time with travel requests arriving dynamically, as shown on the right of Fig. 1.

Each request i can be defined as a tuple $(o_i, d_i, te_i, tl_i, f_i)$, where o_i represents the origin; d_i is the destination; te_i is the earliest departure time; tl_i is the latest departure time; f_i is the profit that request i brings to the operator and equals the order price deducting the travel cost from o_i to d_i . It is assumed that if the operator can dispatch an empty aTaxi to o_i before tl_i , then request i can be served. Otherwise, the traveler will leave the system to seek for another transportation mode or change the travel plan. If i is an immediate request, then the earliest departure time, te_i , is the traveler's request time, and the length of departure time window ($tl_i - te_i$) is the traveler's maximum tolerable waiting time. If i is a predicted request, then the departure time window $[te_i, tl_i]$ represents the time period during which travel requests may occur. If the predicted request is assigned to an aTaxi, then it means that the aTaxi will rebalance to o_i before tl_i , waiting for the emergence of travel requests nearby. In addition, a different attitude can be adopted for prediction because of its error, full trust, or partial trust. Parameter $a \in [0, 1]$ is therefore set to indicate the believability of prediction, which also functions as the discount of predicted request profit, as shown in Eq. (12).

The route of each aTaxi is divided into two parts, i.e., fixed route and alterable route. The fixed route is bound to be executed without being disturbed by the dispatcher at a later decision-making point. The alterable route, which may not be completely executed, will be updated at each decision-making point afterwards. It is assumed that once an immediate request is assigned to an aTaxi, the assignment is fixed, and the route of serving the immediate request is added to the fixed route of the assigned aTaxi. The

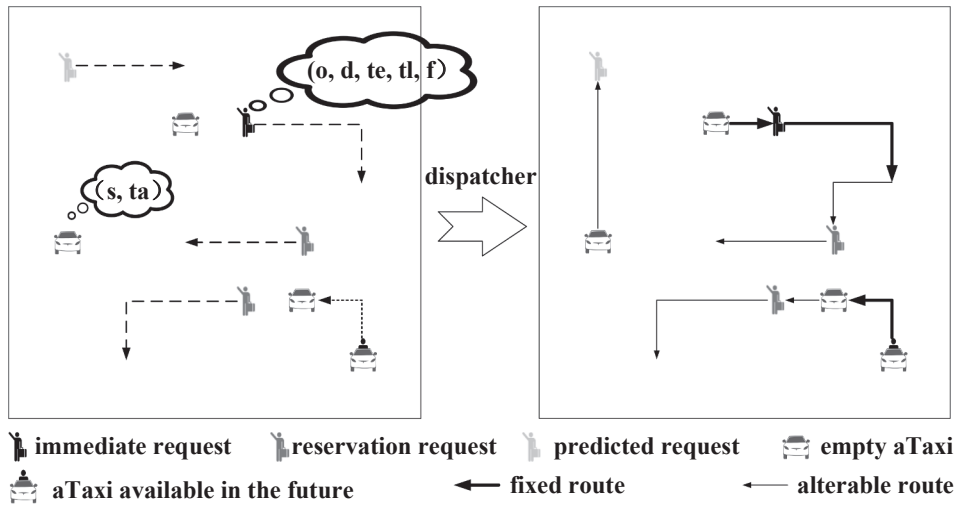


Fig. 1. Problem statement.

assigned aTaxi for the reservation request is allowed to be replaced by another aTaxi, and the en-route rebalancing aTaxi remains available for reassignment to other requests. In other words, the route of serving the reservation requests and predicted requests is an alterable route, as shown on the right of Fig. 1. In addition, the route of serving the reservation requests in an alterable route will be transformed into a fixed route when the departure time of reservation requests is approached.

The status of each aTaxi k can be defined as a tuple (ta_k, s_k) , where ta_k represents the time when aTaxi k becomes available (referred to as available time), and s_k is the location where k becomes available (referred to as available location). For an idle aTaxi or a rebalancing aTaxi, its fixed route is null, and ta and s are the current time and current location of the aTaxi, respectively. For a picking-up aTaxi or a dropping-off aTaxi, ta and s are the estimated time and location of the aTaxi when it finishes its fixed route, respectively. Similar to the aTaxi shown in the lower right corner of Fig. 1, its available time is the estimated drop-off time, and its available location is the destination of the traveler in the aTaxi.

Based on the requests and aTaxi status, the dispatcher will assign travel requests to aTaxis and plan routes for them. It is noted that the focus is not on the actual physical aTaxi route. “Route” in this paper pertains to a sequence of pickup locations, drop-off locations, and rebalance locations. It is represented by a request chain, where an immediate request or a reservation request represents pickup and drop-off locations, and a predicted request represents rebalance locations.

4. Dispatching strategy for aTaxis in single request mode

As groundwork for the centralized dispatcher in Section 5, this section presents an aTaxi dispatching strategy in single request mode where travelers only request immediate rides.

4.1. Planning horizon

To handle the dynamic aspect (dynamic arrival of travel requests and dynamic change in aTaxi status), the rolling horizon procedure is used to repeatedly resolve the problem at an updated interval, Δt . As illustrated in Fig. 2, the whole day is discretized into a sequence of decision-making points, and the planning horizon is divided into short-term and long-term horizons. Requests are categorized into short-term requests of which the earliest departure times are within the short-term horizon and long-term requests of which the earliest departure times are within the long-term horizon. The planned route is divided into two parts: short-term route whose scheduled time is within the short-term horizon and long-term route whose scheduled time is within the long-term horizon. This section only focuses on the assignment of short-term requests and short-term route plans. At the decision-making point, t , the dispatcher collects immediate requests, R_t , that arrived during the time period $[t - \Delta t, t]$ and predicted requests, R_p , whose departure time windows are within $[t, t + T_{hor}]$ and assigns them to aTaxis V , which is available for assignment within the short-term horizon. When the next decision-making point, $t + \Delta t$, arrives, the information and plan routes for the aTaxi will be updated again.

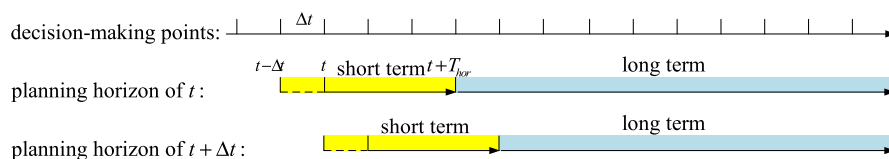


Fig. 2. Illustration of planning horizon.

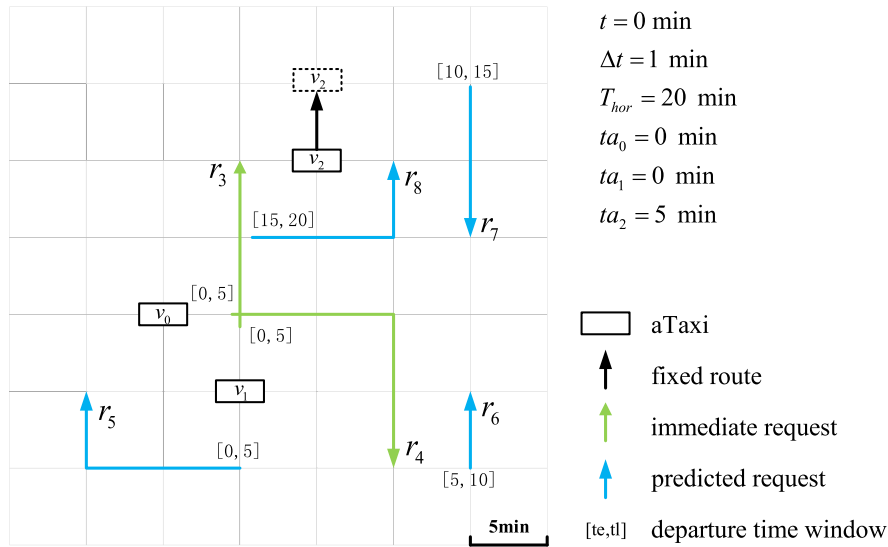


Fig. 3. Request and aTaxi information at 0 min.

Fig. 3 shows a small case, referred to as Example 0, where the travel time in each road section is 5 min, and the short-term horizon length is 20 min. At decision-making point $t = 0$ min, there are three aTaxis in the service area, including idle aTaxis, v_0 and v_1 , and an occupied aTaxi, v_2 , which will drop off travelers at $t = 5$ min. From the time of the last decision-making point to the present, two travelers request immediate rides (r_3 , r_4), and there are four short-term predicted requests (r_5 , r_6 , r_7 , r_8). The dispatcher input at decision-making point $t = 0$ min thus contains $V = \{v_0, v_1, v_2\}$, $R_C = \{r_3, r_4\}$, and $R_P = \{r_5, r_6, r_7, r_8\}$.

4.2. ATaxi-request network

The aTaxi dispatching problem is formulated as a network flow model based on the aTaxi-request network, which draws certain aspects from the method of Vazifteh et al. (2018). Vazifteh et al. (2018) established the vehicle-shareability network where all trips in a whole day are modeled as nodes, and two nodes are linked if the trips represented by these two nodes can be consecutively completed by the same vehicle. Based on the vehicle-shareability network, Vazifteh et al. (2018) modeled the minimum fleet problem as a graph theory problem for finding minimum disjoint paths to cover all nodes. This method is expanded to real-time aTaxi dispatching in the paper. A similar network, aTaxi-request network, is constructed as (N, E) , where $N = V \cup R \cup \{S\}$ and $E = E_{VR} \cup E_{RR} \cup E_{VS} \cup E_{RS}$, as listed in Table 3. Compared with the vehicle-shareability network of Vazifteh et al. (2018), three types of node and four types of arc are incorporated in this paper. ATaxi node is added to solve an aTaxi-request assignment problem. Request node can be classified into two categories: immediate request node (to be assigned) and predicted request node (guides vehicles to rebalance). Sink node represents the end of short-term horizon. Four types of arcs are reflected as four vehicle behaviors.

In view of the departure time window of request, the conditions for establishing the pickup arc (k, i) and relocation arc (i, j) are as follows.

$$ta_k + t(s_k, o_i) \leq tl_i \quad (1)$$

$$tl_i + t(o_i, d_i) + t(d_i, o_j) \leq tl_j \quad (2)$$

Table 3
Types of nodes and arcs in aTaxi-request network.

Type	Notation	Meaning
Node		
ATaxi node	$k \in V$	Represents an aTaxi, k
Request node	$i \in R$	Represents a request, i
Sink node	S	Represents end of short-term horizon
Arc		
Pickup arc	$(k, i) \in E_{VR}$	Represents that aTaxi k picks up and delivers traveler i
Relocation arc	$(i, j) \in E_{RR}$	Represents that aTaxi is relocated to serve traveler j after dropping off i
Waiting arc	$(k, S) \in E_{VS}$	Means that aTaxi k remains inactive after it becomes available until the end of short-term horizon
Ending arc	$(i, S) \in E_{RS}$	Means that aTaxi remains inactive after dropping off i until the end of short-term horizon, or time is beyond short-term horizon after dropping off i .

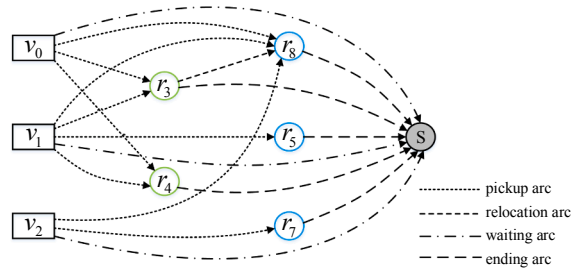


Fig. 4. aTaxi-request network at 0 min.

If k and i satisfy Inequality (1), then this indicates that traveler i can be picked up by aTaxi k before the his/her latest departure time, and nodes k and i will be linked. If i and j satisfy Inequality (2), then this indicates that requests i and j can be consecutively completed by a single aTaxi with no ridesharing, and nodes i and j will be linked. It should be noted that real requests (immediate requests and reservation requests) and virtual requests (predicted requests) are treated in the same way when establishing network and building model except their profit (as discussed in Section 3). On the one hand, Inequalities (1) (2) are also the conditions for establishing arcs to predicted requests, because it is ineffective to dispatch vehicle to the origin of predicted request after the time period during which the request may occur. On the other hand, it is assumed that completing a predicted request also represents that vehicle travels from its origin to its destination like a real request when establishing network and building model, which means that vehicle reserves the capacity for serving the requests that may occur in the future. But vehicle only need to rebalance to the origin of predicted request when executing. Therefore, the pickup arc or relocation arc to predicted request actually means that vehicle will rebalance to the origin of predicted request. For example, the pickup arc $v_0 \rightarrow r_8$ in Fig. 4 means that v_0 will rebalance to the origin of r_8 . The relocation arc $r_3 \rightarrow r_8$ means that aTaxi will rebalance to the origin of r_8 after dropping off r_3 .

Based on the foregoing conditions, the aTaxi-request network for Example 0 is built, as shown in Fig. 4.

In an attempt to maximize the total revenue, the profit on pickup arc (k, i) is defined as the profit from serving traveler i subtracting the travel cost of picking up, and the profit on relocating arc (i, j) as the profit from serving traveler j subtracting the travel cost of relocation. The profits on the waiting and ending arcs are both zero.

$$\begin{aligned} p_{ki} &= f_i - c(s_k, o_i) \quad \forall (k, i) \in E_{VR} & p_{kS} &= 0 \quad \forall (k, S) \in E_{VS} \\ p_{ij} &= f_j - c(d_i, o_j) \quad \forall (i, j) \in E_{RR} & p_{iS} &= 0 \quad \forall (i, S) \in E_{RS} \end{aligned} \quad (3)$$

4.3. Network flow model

In the aTaxi-request network, the path from aTaxi node k to sink node S represents the short-term route of k . If a predicted request is assigned to an aTaxi, then this means that the aTaxi will rebalance to the origin of the predicted request, waiting for the occurrence of a future request nearby. For example, the path $v_1 \rightarrow r_3 \rightarrow r_8 \rightarrow S$ in Fig. 4 reflects the planned route of v_1 between 0 and 20 min. This means that v_1 will serve r_3 first and thereafter be relocated to the origin of r_8 where it waits for a future request nearby. The focus is thus shifted to the selection of a path from each aTaxi node to sink node S in the aTaxi-request network with all paths disjoint except on the sink node because a request cannot be served by more than one aTaxi. Based on the aTaxi-request network, the problem is modeled as a maximum cost flow problem.

$$\max \sum_{(k,i) \in E_{VR}} p_{ki} x_{ki} + \sum_{(i,j) \in E_{RR}} p_{ij} x_{ij} \quad (4)$$

$$\sum_{i \in R} x_{ki} + x_{kS} = 1 \quad \forall k \in V \quad (5)$$

$$\sum_{k \in V} x_{kj} + \sum_{i \in R} x_{ij} = \sum_{i \in R} x_{ji} + x_{jS} \quad \forall j \in R \quad (6)$$

$$\sum_{k \in V} x_{kS} + \sum_{i \in R} x_{iS} = n \quad (7)$$

$$\sum_{k \in V} x_{kj} + \sum_{i \in R} x_{ij} \leq 1 \quad \forall j \in R \quad (8)$$

$$\begin{aligned} x_{ki} &= \{0, 1\} \quad \forall (k, i) \in E_{VR} & x_{kS} &= \{0, 1\} \quad \forall (k, S) \in E_{VS} \\ x_{ij} &= \{0, 1\} \quad \forall (i, j) \in E_{RR} & x_{iS} &= \{0, 1\} \quad \forall (i, S) \in E_{RS} \end{aligned} \quad (9)$$

Eq. (9) shows the binary constraints of decision variables, where x_{ki} , x_{kS} , x_{ij} , and x_{iS} represent the flow on the four types of arcs. Eq. (4) is the objective function that aims to maximize the total revenue in the short-term horizon. Constraints (5)–(7) are flow balance constraints for source nodes V , intermediate nodes R , and sink node S , respectively. Constraint (5) also means that the aTaxi

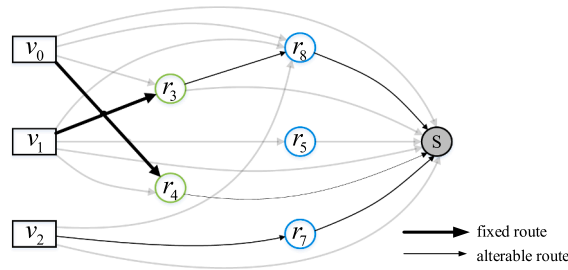


Fig. 5. Planned route for each aTaxi at 0 min.

can choose one of the behaviors (waiting, serving immediate request, or rebalancing to the origin of predicted request) when it becomes available. Constraint (6) means that the aTaxi can perform other activities after serving request j . Constraint (8) shows that a request can be served by no more than one aTaxi.

The solution of above model recommends a route to each aTaxi. Fig. 5 shows the recommended routes of Example 0: v_0 will immediately pick up and deliver r_4 ; v_1 will first serves r_3 and then rebalance to the origin of r_8 , waiting for the emergence of travel requests nearby; v_2 will rebalance to the origin of r_7 after dropping off. Then, the fixed route and alterable route of aTaxi will be updated by recommended route. If there are one or more immediate requests in recommended route, the last immediate request in recommended route, referred to as r_i , is regarded as the demarcation point. The part of route before it (including it) is added into fixed route, and the part of route after it acts as new alterable route with original alterable route removed. Under most circumstances, r_i is also the first request in recommended route. There are one or more requests before r_i in very few cases where the distance of the requests before r_i is too short and their origins and destinations are too close. After route dividing, the available time and the available location of aTaxi are updated as the estimated time and location of completing r_i , respectively. Thus, the fixed route will not be disturbed at later decision making points; If there is no immediate request in recommended route, no requests will be added into fixed route, and the alterable route is updated as the recommended route. The available time and location is still estimated time and location of completing fixed route, or current time and location if fixed route is null, respectively. In route $v_1 \rightarrow r_3 \rightarrow r_8 \rightarrow S$, “ $v_1 \rightarrow r_3$ ” is added into the fixed route of v_1 , and the alterable route of v_1 is updated as “ $\rightarrow r_8 \rightarrow S$ ” as stated above. The available time and location of v_1 are updated accordingly as $ta_1 = 15$ and $s_1 = (4, 5)$, respectively, to ensure that the fixed route will not be disturbed in a later decision-making point. Similarly, the available time and location of v_0 are updated as $ta_0 = 25$ and $s_0 = (5, 1)$, respectively, and v_2 remains constant in the available time and location.

aTaxi travels following the planned route. The first request in planned route, referred to as r_f , is current task of aTaxi. r_f is the first request in fixed route if fixed route is not null, or the first request in alterable route otherwise. If r_f is an immediate request, the current task is picking up and delivering the passenger. If r_f is an predicted request, the current task is rebalancing to the origin of r_f . Once the task is completed, it will be removed from fixed route.

5. Dispatching strategy for aTaxi in hybrid request mode

In railway transportation, the decentralized autonomous centralized traffic control system not only realizes centralized control of trains but also allows stations to control signals and train routes autonomously in the decentralized autonomous mode. Drawing on this idea, a centralized dispatcher and decentralized autonomous dispatchers are designed to solve the aTaxi dispatching problem in hybrid request mode where immediate requests and reservation requests coexist.

The centralized dispatcher recommends optimal short-term routes to all aTaxi by central planning in a way similar to that of the dispatching strategy in single request mode. The decentralized autonomous dispatcher, which is loaded on each aTaxi, allows the aTaxi to plan its own long-term route and keep its route feasible. On the one hand, as an autonomous agent with communication and computing capabilities, the aTaxi is capable of managing its own route, which makes this proposed dispatching strategy technically possible. On the other hand, it is unnecessary for a long-term reservation request (with a departure time window that is distant from its request time) to be reassigned by central planning when the future is vague. Its incorporation into central planning can wait until it enters the short-term horizon. Before that, the decentralized autonomous dispatchers can maintain the fleet capacity to serve the long-term reservation request. The centralized dispatcher’s workload is thus reduced.

5.1. Centralized dispatcher

The centralized dispatcher adds the assignment of short-term reservation requests based on the dispatching strategy presented in Section 4. It similarly collects travel requests, vehicle status information, and planning route for each aTaxi at an updated interval. The travel request input is expanded to $R = R_I \cup R_P \cup R_{Rn1} \cup R_{Ra1}$, where R_{Rn1} collects newly arrived short-term reservation requests, and R_{Ra1} collects the short-term reservation requests previously accepted by the dispatcher.

The difference between request time and expected departure time of the reservation request generates two sub-problems: how to respond to a traveler’s reservation request (i.e., whether or not to accept the request) when future information is lacking, and how to guarantee that accepted reservation requests can be served. For the second sub-problem, Constraint (8) is rewritten as follows.

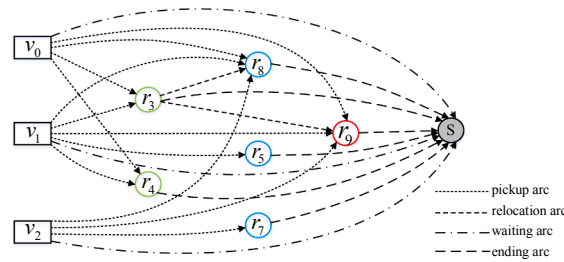


Fig. 6. aTaxi-request network at 0 min.

$$\sum_{k \in V} x_{kj} + \sum_{i \in R} x_{ij} \leq 1 \quad \forall j \in R_I \cup R_P \cup R_{Rn1} \quad (10)$$

$$\sum_{k \in V} x_{kj} + \sum_{i \in R} x_{ij} = 1 \quad \forall j \in R_{Ra1} \quad (11)$$

The model is therefore modified as Eqs. (4)–(7) and (9)–(11). It remains a network flow model and can be solved by state-of-the-art solvers. For the first sub-problem, if the newly arrived reservation request, $i \in R_{Rn1}$, is assigned to an aTaxi in the solution of the modified model (indicating that the fleet has capacity to serve i and it is worthwhile to do so), then the traveler can be informed that the request is accepted, and request i is thereafter transferred from R_{Rn1} to R_{Ra1} . Otherwise, i is rejected.

After acquiring the recommended route by the modified model, the short-term route is updated in the same way referred in Section 4.3. In addition, if the first request in alterable route (referred to as r_{af}) is a reservation request, we will check whether it is “close” to fixed route. If the estimated pickup time of r_{af} is beyond its earliest departure time (i.e. $ta + t(s, o_{af}) \geq te_{af}$), r_{af} is called “close” to fixed route. The idle time length from the available time of aTaxi to picking up r_{af} is short when r_{af} is “close” to fixed route, so the assignment is relatively good and unnecessary to be changed. Therefore, if r_{af} is “close” to the fixed route, it will be moved to the fixed route from the alterable route, and the available time and available location of the aTaxi will be updated accordingly. This way can avoid troubling passenger with frequent reassignment when the time is approaching departure time.

5.2. Centralized dispatcher example

Example 1 is designed to illustrate the centralized dispatcher’s work, where there is a reservation request, $r_9(o_9 = (2, 5), d_9 = (0, 5), te_9 = 16, tl_9 = 21, f_9 = 10)$, which arrives as an additional request at 0 min in Example 0, as shown in Fig. 3. With the request and aTaxi information that have been acquired, the aTaxi-request network is built as Fig. 6.

Solving the above model can yield the result shown in Fig. 7, where reservation request r_9 is assigned to aTaxi v_1 . The centralized dispatcher will therefore respond to the traveler that r_9 is accepted; r_9 is then transferred from R_{Rn1} to R_{Ra1} . Constraint (11) is added to r_9 at each decision-making point henceforth to ensure that r_9 can be served regardless of how the planned route of each vehicle changes. In the current plan, there is at least one aTaxi that can pick up r_9 at the next decision-making point so that the model built at the next decision-making point definitely has a feasible solution without violating Constraint (11) because r_9 can be served by v_1 . By analogy, it is found that the centralized dispatcher can guarantee that the model has a feasible solution at any decision-making point.

5.3. Decentralized autonomous dispatcher and its algorithm

By taking advantage of the aTaxi autonomy, the decentralized autonomous dispatcher on the aTaxi receives the short-term route recommended by the centralized dispatcher and determines the real route to be executed. The decentralized autonomous dispatcher on the aTaxi monitors and responds to long-term reservation requests nearby and plans long-term route for the aTaxi. It is unnecessary to make an optimal assignment when long-term requests arrive because these will become short-term requests over time and will be reassigned by the centralized dispatcher. A heuristic method in the decentralized autonomous dispatcher is used to assign long-term reservation requests, thereby improving computational efficiency. It is necessary for the decentralized autonomous

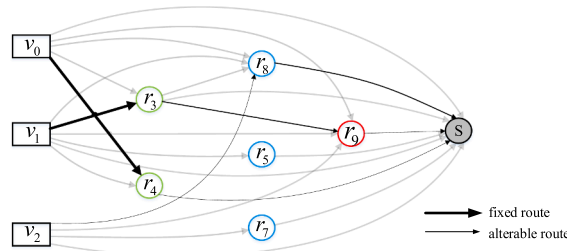


Fig. 7. Short-term route for each aTaxi at 0 min.

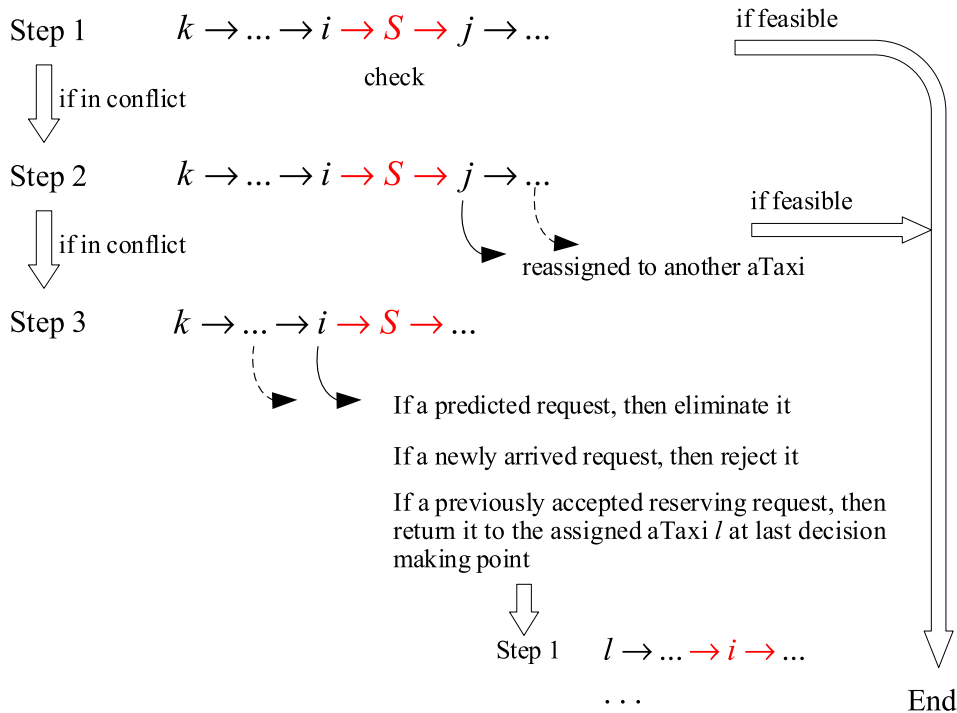


Fig. 8. Procedure for checking and adjusting planned route.

dispatcher on each aTaxi to check and adjust the route to maintain the connection between the two routes feasible after receiving the short-term route recommended by the centralized dispatcher. This is because central planning does not consider potential conflicts between short-term and long-term routes. The specific work of a decentralized autonomous dispatcher is illustrated as follows:

(1) Assign newly arrived long-term reservation requests

The decentralized autonomous dispatcher on each aTaxi constantly monitors the occurrence of long-term reservation requests nearby. When a traveler requests a long-term reservation, $j \in R_{Rn2}$, its information will be immediately captured by the nearest aTaxi. The decentralized autonomous dispatcher will examine whether serving j can be inserted into its alterable route without time conflict according to Inequalities (1) and (2). If it can be inserted, then j will be assigned to it. Otherwise, the decentralized autonomous dispatcher will transmit the information of j to another nearby aTaxi, which will conduct the same examination. If j can be accepted, then the transmission ends. Otherwise, j will continue to be transmitted to other aTaxis until j can be accepted by one aTaxi. If no aTaxi in the fleet can accept j after examination, j will be rejected.

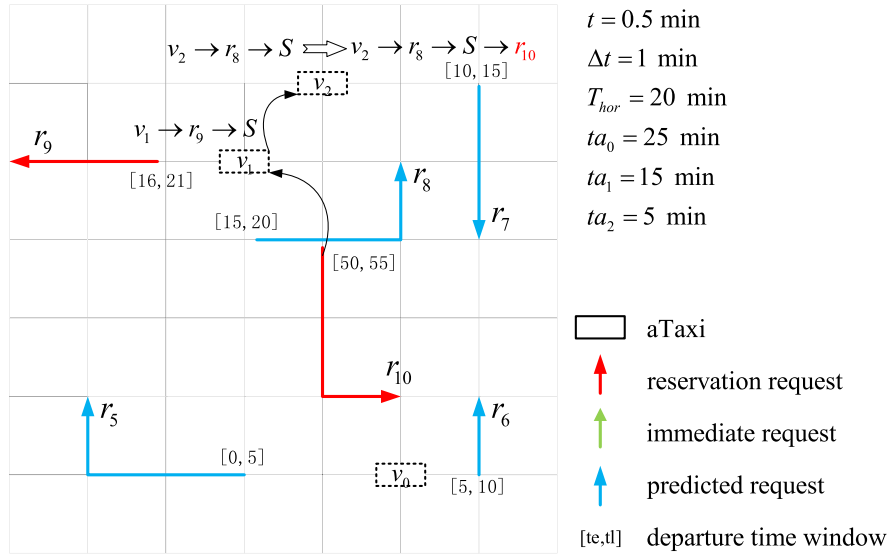
(2) Check and adjust the route after central planning

Consider aTaxi k whose planned route is updated as $k \rightarrow \dots \rightarrow i \rightarrow S \rightarrow j \rightarrow \dots$ (after central planning is performed by the centralized dispatcher) as an example to illustrate the approach to maintain the connection between short-term and long-term routes feasible (see Fig. 8).

Step 1: The decentralized autonomous dispatcher on k will check whether it can serve i and j consecutively according to Inequality (2) without considering S because S is a virtual node. If they can be served, then the planned route is feasible, and it is unnecessary to execute the succeeding two steps.

Step 2: If there is conflict between i and j , then the decentralized autonomous dispatcher will reassign j to another aTaxi in the same way as the newly arrived reservation request is assigned. If j can be accepted by another aTaxi, then it will be eliminated from the planned route of k ; another check is thereafter performed. If the planned route remains infeasible, then the reservation requests behind j are reassigned successively until the reassignment fails or the planned route becomes feasible. If the planned route becomes feasible, then the adjustment ends. Otherwise, proceed to execute Step 3.

Step 3: If the planned route remains infeasible, then the decentralized autonomous dispatcher will eliminate i from the planned route. If i is a predicted request, then there is no scruple at all. If i is a newly arrived request, then the traveler is informed that i is rejected. If i is a previously accepted reservation request, then it will be returned to the assigned aTaxi l at the last decision-making point. ATaxi l will check and adjust its planned route in the same way until its planned route becomes feasible. If the planned route

Fig. 9. Assignment of r_{10} at 0.5 min.

becomes feasible after eliminating i , the adjustment ends. Otherwise, the requests before i in the planned route are eliminated in the same way until the planned route becomes feasible.

Through checks and adjustments, all planned routes will be feasible because the worst scenario is that they revert to the planned routes before the central planning by the centralized dispatcher, and all newly arrived requests are rejected.

(3) Transfer the requests in R_{Ra2} to the centralized dispatcher

Although the requests in R_{Ra2} are presently within the long-term horizon, they will successively come into the short-term horizon over time. The decentralized autonomous dispatcher will check the earliest departure time of the requests in R_{Ra2} before the central planning by the centralized dispatcher at each decision-making point. Once the request, $j \in R_{Ra2}$, is covered by the short-term horizon, it is transferred from R_{Ra2} to set R_{Ra1} as the model input of the centralized dispatcher.

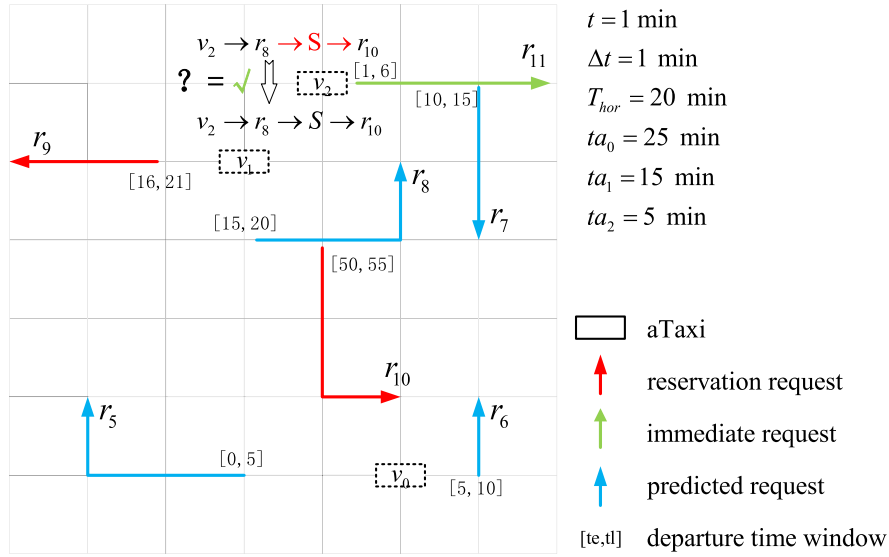
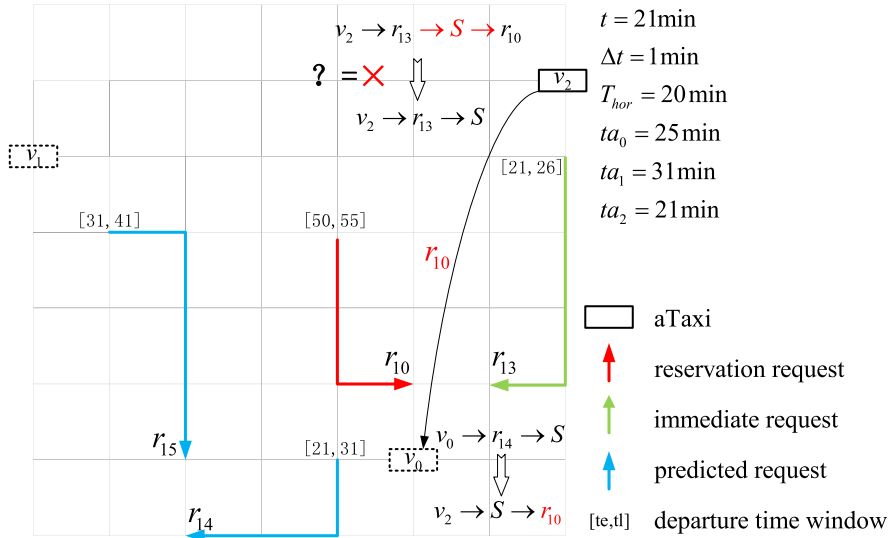
5.4. Decentralized autonomous dispatcher example

Example 2 is designed to illustrate the work of decentralized autonomous dispatchers, where a traveler requests a reservation r_{10} ($o_{10} = (4, 4)$, $d_{10} = (5, 2)$, $te_{10} = 50$, $tl_{10} = 55$, $f_{10} = 15$) at 0.5 min based on Example 1 illustrated in Section 5.2. When r_{10} occurs, the decentralized autonomous dispatcher on v_1 immediately captures and checks whether serving it can be inserted into its planned route. In view of the conflict between r_9 and r_{10} , the latter is transmitted to aTaxi v_2 , which finds that it can accept r_{10} . Reservation r_{10} is thus assigned to v_2 , and the planned route of v_2 is updated as $v_2 \rightarrow r_8 \rightarrow S \rightarrow r_{10}$. The following figures only show the available location and planned route after the available time of aTaxi because the estimated status of the aTaxi after completing the fixed route is the input of route planning (see Fig. 9).

At the decision-making point, $t = 1$ min, the centralized dispatcher collects immediate request r_{11} ($o_{11} = (4, 6)$, $d_{11} = (2, 6)$, $te_{11} = 1$, $tl_{11} = 6$, $f_{11} = 10$), and the predicted requests remain unchanged. After the central planning by the centralized dispatcher, the planned route of v_2 is updated as $v_2 \rightarrow r_{11} \rightarrow S \rightarrow r_{10}$. The decentralized autonomous dispatcher on v_2 checks its planned route and finds it feasible. It is therefore unnecessary to adjust the planned route, and r_{11} is accepted (see Fig. 10).

Similarly, the short-term route of v_2 does not conflict with r_{10} at any decision-making point between 2 and 20 min. At 21 min, the centralized dispatcher collects requests and aTaxi information, as shown in Fig. 11, and reroutes v_2 as $v_2 \rightarrow r_{13} \rightarrow S \rightarrow r_{10}$. The decentralized autonomous dispatcher on v_2 finds the conflict between r_{13} and r_{10} and consequently attempts to reassign r_{10} to another aTaxi. The decentralized autonomous dispatcher on v_0 finds that r_{10} can be inserted into its planned route by eliminating predicted request r_{14} . Reservation r_{10} is therefore reassigned to v_0 , and the planned route of v_0 is updated as $v_0 \rightarrow S \rightarrow r_{10}$. The planned route of v_2 accordingly resumes its original feasibility ($v_2 \rightarrow r_{13} \rightarrow S$).

As can be observed from the foregoing, the decentralized autonomous dispatchers can ensure that there is an aTaxi that can serve r_{10} in the fleet all the time until r_{10} comes into the short-term horizon at 30 min. By this time, r_{10} will have been transferred from R_{Ra2} to R_{Ra1} . The centralized dispatcher will be responsible for the reassignment of r_{10} until it is served.

Fig. 10. Route checking and adjustment of v_2 at 1 min.Fig. 11. Route checking and adjustment of v_2 at 21 min.

6. Experiments

6.1. Experimental settings

An experimental setting similar to that reported in literature (Alonso-Moraa et al., 2017) is adopted, and the aTaxi operation performance in Manhattan, New York, is investigated.

6.1.1. Road network

A road network of Manhattan, obtained from the spatial data offered by openstreetmap.org, contains 5023 nodes and 10 137 directed links as shown in Fig. 12. The streets of Manhattan are filtered, and those in the following classifications are selected: primary, secondary, tertiary, residential, unclassified, road, and living street. The roads are thereafter split into several segments by street intersections so that each road segment connects two adjacent intersections without containing any intersections except at beginning and end of the segment. The directions of road segments are thereafter identified because many streets in Manhattan are one-way. Finally, intersections and road segments are extracted to construct the road topology, where nodes represent intersections, and directed links represent road segments. It is noted that the number of nodes and directed links are more than those employed in Alonso-Moraa et al. (2017) (4091 nodes and 9452 directed links), because the intersections of selected roads (primary, secondary,

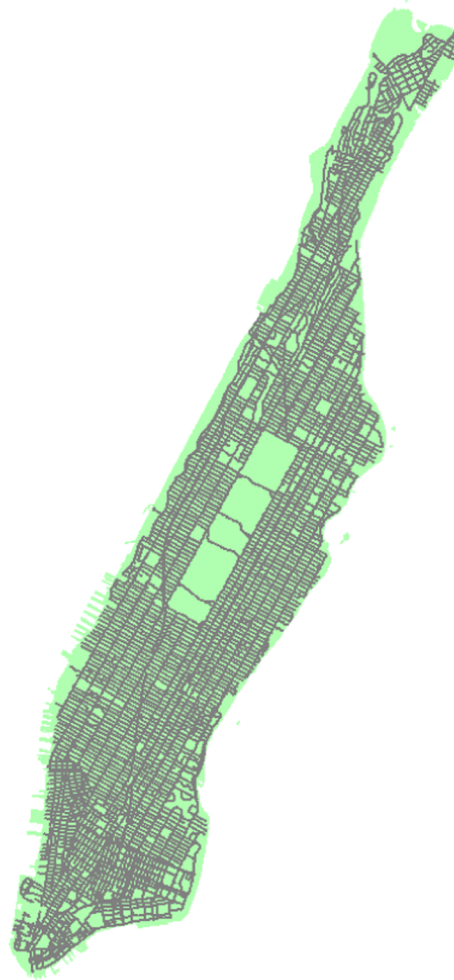


Fig. 12. Road network of Manhattan.

tertiary, residential, unclassified, road, and living street) and discarded roads (footpaths, links, service roads, and so on) that remain can also be pickup or drop-off locations.

It is assumed that the vehicle travels at the same speed in each road: 5.5 m/s between 7 am and 7 pm and 8.5 m/s in the evening. The shortest paths and travel times between any two nodes are computed and stored before formal experiments are conducted. A more realistic speed estimation, such as the method employed by [Alonso-Moraa et al. \(2017\)](#), is possible if necessary.

6.1.2. Request generation

The dataset ([Donovan, 2016](#)) is employed to test the proposed dispatching strategies. This dataset covers four years of taxi operations in New York City. Each row of the data represents a single taxi trip. Each trip includes information on the vehicle permit, vehicle license, vendor ID, rate code, pickup and delivery times, passenger count, trip time, trip distance, and latitude and longitude coordinates for pickup and delivery locations.

The trip data are preprocessed before they are utilized. The loop trips (the trips with the same pickup and delivery locations), trips with unrealistic distance (shorter than 0.3 mi or longer than 13 mi) or unrealistic speed (slower than 0.5 m/s or faster than 30 m/s), are eliminated. The trips where pickup and delivery locations are both within Manhattan are extracted. Then pickup and drop-off locations are matched to the closest nodes in the above road network. The pickup time reported by the trip data is assumed as the earliest departure time. The length of the departure time window of each request is set to 5 min.

A workday is randomly selected for the formal experiments—Tuesday 10/08/2013. Twenty percent of the trip data on this day are selected and divided into two parts according to reservation rate, immediate requests, and reservation requests. The request time of the immediate request is equal to the earliest departure time. The request time of the reservation request is determined by the advance time of reservation (time length between request time and earliest departure time of reservation request), which is sampled with a negative exponential distribution. This distribution is reasonable considering that travelers prefer to reserve an aTaxi near the departure time. When it is time for the request time, the request enters into the system.

The prediction of future requests is always based on historic trips data. For simplicity, the trips in previous Tuesday's (10/01/

2013) trips data is directly used as predicted requests in the experiments. The predicted requests in this paper reflect the immediate requests that may occur in the future, so the proportion of predicted requests used to select from the trips data of 10/01/2013 is set as the proportion of immediate requests in trips data of 10/08/2013. For example, when reservation rate is 20%, immediate requests account for 16% of the trips data in 10/08/2013 because total requests come from 20% of the trips data in 10/08/2013. So we select 16% requests randomly from the trips data in 10/01/2013 as predicted requests after preprocessing. In the central planning of centralized dispatching, the predicted requests with departure time windows in the short-term horizon are employed as input to the proposed model. The profit, which request i brings to the operator, is set as follows:

$$f_i = \frac{A + B \times \text{dist}(o_i, d_i) - P \times \text{dist}(o_i, d_i)}{(A + B \times \text{dist}(o_i, d_i) - P \times \text{dist}(o_i, d_i)) \times a} \quad i \in R/R_p \quad (12)$$

where a represents believability of prediction; P represents the travel cost per kilometer, and the values of A , B , and P are \$2.5, \$1.25/km, and \$0.1/km, respectively.

6.1.3. Vehicle

The experiments are conducted in three fleet sizes: 300, 500, and 700. The initial location of each aTaxi is determined by the warm-up phase. In this phase, the 10/07/2013 trip data are used to simulate the aTaxi operation, and the aTaxi location at the end of this day is recorded as the initial location in the formal experiments.

The vehicle's movement follows the dispatching strategies, and the stored shortest paths between any two nodes. A detailed rebalancing rule is also developed. The rebalancing vehicle does not have to travel to the exact origin of the predicted request. When the vehicle is within 5 min (length of departure time window) away from the origin of the predicted request, the rebalance task is regarded as complete. In this way, not only is the rebalance distance reduced, but the rebalance purpose is also achieved. If a request occurs near the origin of the predicted request, the vehicle can also reach its origin within its departure time window in high probability.

6.1.4. Model setting

The length of the short-term horizon of the proposed dispatching strategies is set to 30 min because most requests can be completed within this time. The updated interval is set to 2 min so that each request can be responded to within 2 min.

6.1.5. Experimental environment

The experiments are realized by Java programming on a PC with i5-7300HQ CPU @ 2.50 GHz and 8-GB memory, and IBM CPLEX is used directly to solve the models of dispatching strategies.

6.2. Experiment 1

Experiment 1 tests the aTaxi dispatching strategy discussed in Section 4. In single request mode, four aTaxi dispatching strategies are compared. Strategy 1 dispatches the nearest idle aTaxi to immediately serve newly arrived requests. Strategy 2 simplifies the dispatching strategy presented in Section 4 without an empty vehicle rebalance, with $R_p = \emptyset$. Strategy 3 is the same as Strategy 2 except that it changes the objective function to maximize the number of travelers that can be served, as shown in Eq. (13). Strategy 4 is the dispatching strategy described in Section 4.

$$\max \sum_{j \in R} \left(\sum_{k \in V} x_{kj} + \sum_{i \in R} x_{ij} \right) \quad (13)$$

The first three strategies assume that idle aTaxis remain in place when they are not assigned to serve travelers. The aTaxi

Table 4

Comparison of four strategies in single request mode.

Strategy	Strategy 1	Strategy 2	Strategy 3	Strategy 4
Fleet size	500	500	500	500
Number of generated requests	48,592	48,592	48,592	48,592
Number of served requests	33,332	29,751	36,696	46,377
Completion rate of requests (%)	68.596	61.226	75.519	95.442
Average waiting time (min)	3.937	4.171	4.531	3.735
Total revenue(\$)	192673.41	183253.17	211882.21	270507.76
Total travel distance (km)	96853.631	96097.196	106930.939	137122.215
Total pickup distance (km)	20389.713	16361.03	28287.715	26561.183
Total rebalance distance (km)	0	0	0	4691.73
Travel distance per trip (km)	2.906	3.23	2.914	2.957
Pickup distance per trip (km)	0.612	0.55	0.771	0.573
Rebalance distance per trip (km)	0	0	0	0.101
Average runtime (s)	0.003	0.013	0.014	10.264

operation is simulated the whole day in the four dispatching strategies. Their operational performances are listed in Table 4.

As can be observed from the above results, the total revenue and the number of served requests of Strategy 2 are lower than those of other strategies. This reflects the myopic defect of Strategy 2. It is found that the profit acquired by serving a long-distance request is lower than the total profits gained by serving multiple short-distance requests with the same total travel distance computed according to Eq. (12). Strategy 2 cannot predict multiple short-distance requests that may occur in the future, and it prefers to serve current long-distance requests because of its objective function, as indicated by its higher travel distance per trip of 3.23 km. Serving long distance requests means a long-time occupancy of aTaxi. If short-distance requests occur later, the occupied aTaxi cannot serve them. The myopic defect results in the inadequate performance of Strategy 2.

Although Strategy 3 performs better among the first three strategies in terms of total revenue and completion rate of requests, its average waiting time and pickup distance is highest.

Strategy 4 achieves the best performance in both total revenue and service quality. In terms of economic benefits, its total revenue is \$58625.55 higher than Strategy 3 at \$117.25 per vehicle. In terms of service quality, it serves most requests (only 4.558% of requests is rejected) with shortest passenger waiting time (3.735 min). As a result of the incorporation of predicted requests, however, the number of input requests of Strategy 4 is approximately 15 times that of other strategies, resulting in the longest runtime. Compared with the updated interval of 2 min, the runtime of 10.264 s is acceptable.

6.3. Experiment 2

The centralized dispatching strategy, where all requests (including long-term reservation requests) are assigned centrally, is designed for comparison with the proposed dispatching strategy. In the centralized dispatching strategy, the request input is set as $R = R_I \cup R_P \cup R_{Rn1} \cup R_{Ra1} \cup R_{Rn2} \cup R_{Ra2}$, and Constraints (10) and (11) are rewritten as Constraints (14) and (15). The model of centralized dispatching strategy is represented by Eqs. (4)–(7), (9), and (14)–(15).

$$\sum_{k \in V} x_{kj} + \sum_{i \in R} x_{ij} \leq 1 \quad \forall j \in R_I \cup R_P \cup R_{Rn1} \cup R_{Rn2} \quad (14)$$

$$\sum_{k \in V} x_{kj} + \sum_{i \in R} x_{ij} = 1 \quad \forall j \in R_{Ra1} \cup R_{Ra2} \quad (15)$$

Experiment 2 compares the centralized dispatching strategy (Cen) and decentralized autonomous and centralized dispatching strategy (De&Cen) in hybrid request mode where reservation requests account for 40% of the total requests. Twelve percent of last Tuesday's (10/01/2013) trip data are employed as predicted requests because immediate requests come from 12% of the 10/08/2013 trip data. The Cen and De&Cen are tested with the same requests and fleet. The operational performances of the two strategies are listed in Table 5.

As listed in Table 5, there is a slight difference between the two strategies in operational performances. De&Cen performs similar to Cen not only in economic benefits but also in service quality (i.e. similar completion rate of requests and similar average waiting time), probably because long-term reservation requests may disturb the current decision-making of Cen. Long-term reservation requests only account for a part of long-term requests; however, Cen considers them as all requests in the long-term horizon. Compared to De&Cen, which does not incorporate long-term requests into central planning, Cen does not achieve better operational performance.

The biggest problem with Cen is in terms of computational efficiency. The runtime of Cen is overly long for application in real-time vehicle dispatching, as shown in Fig. 13. The average runtime of Cen is approximately eight times that of De&Cen. In the Cen dispatching strategy, the long-term reservation request has been the model input over a considerable time. This is because its departure time window is distant from the present, and Cen always adjusts its assignment until departure time. As a result, long-term reservation requests utilize a considerable amount of the computing resources in the Cen dispatching strategy. In the De&Cen dispatching strategy, however, the long-term reservation request does not use the computing resources of central planning before it

Table 5
Comparison of Cen and De&Cen in hybrid request mode.

Strategy	Cen	De&Cen
Fleet size	500	500
Number of generated requests	48,592	48,592
Number of served requests	46,339	46,395
Completion rate of requests (%)	95.363	95.479
Average waiting time (min)	3.465	3.451
Total revenue(\$)	270067.93	270307.28
Total travel distance (km)	136870.814	136989.311
Total pickup distance (km)	30040.755	30525.788
Total rebalance distance (km)	1772.37	1653.48
Travel distance per trip (km)	2.954	2.953
Pickup distance per trip (km)	0.648	0.658
Rebalance distance per trip (km)	0.038	0.036
Average runtime (s)	106.071	13.014

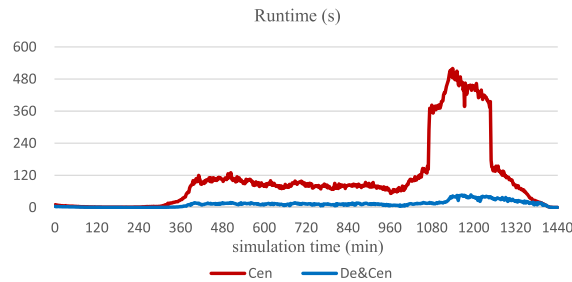


Fig. 13. Runtime comparison between Cen and De&Cen.

enters into the short-term horizon. Decentralized autonomous dispatchers improve computational efficiency.

6.4. Experiment 3

Experiment 3 tests the robustness of the proposed dispatching strategies with respect to errors in predictions in a fleet size of 500. The proposed dispatching strategies require predicted requests with detailed information. The method to get predicted requests (i.e. directly select from someday before) is simple and rough, and the error between predicted requests and immediate requests during the same time period is obviously great. However, the operational performance is also very good as shown in Table 4, which reflects the robustness of the dispatching strategy. In addition, to represent three scenes with different prediction errors, three types of predicted requests are designed, which are selected from trips data of 10/01/2013 with different proportion. The “accurate” proportion used to get “accurate” predicted requests is set as the proportion of immediate requests in trips data of 10/08/2013. The “excessive” proportion is set as 1.5 times of the “accurate” proportion. The “insufficient” proportion is set as 0.5 times of the “accurate” proportion. We select the requests randomly from the trips data of 10/01/2013 with the “accurate” proportion as the “accurate” predicted requests, the requests with the “excessive” proportion as the “excessive” predicted requests, and the requests with the “insufficient” proportion as the “insufficient” predicted requests. For example, when reservation rate is 20%, we select 16% requests from the trips data in 10/01/2013 as “accurate” predicted requests, 24% of the trips data as “excessive” predicted requests, and 8% of the trips data as “insufficient” predicted requests. The dispatching strategy is firstly tested with these three predicted requests in single request mode.

As shown in Fig. 14, the operational performance is less sensitive to the believability of predicted requests when the believability is greater than 0 based on the rebalancing rule. On the one hand, rebalancing vehicle keeps it available for reassignment when immediate request or reservation request is posted. Although the vehicles are more probable to be assigned predicted requests with increasing believability, this situation does not prevent vehicles from serving passengers. On the other hand, rebalancing vehicle is only required to travel to somewhere approximately 5 min away from the origin of predicted request. The rebalancing task can therefore be finished considerably quick, and the unoccupied distance rate does not increase excessively. In exactly the same way, predicted requests that are “excessive” only have a slight influence on the total revenue and completion rate of requests although they induce a slight increase in unoccupied distance rate. This can be observed from the red and green lines that are very close in Fig. 14(a) and (b). When predicted requests are fewer than “accurate” predicted requests, however, the rebalance is not sufficient, and a considerable gap between the supply and demand of vehicles still exists in some areas. The total revenue and completion rate with “insufficient” predicted requests are worse.

Next, the robustness of the dispatching strategy in hybrid request mode is also tested with the three predicted requests.

Similarly, the total revenue and completion rate of requests in “excessive” predicted requests are considerably close to those in “accurate” predicted requests in hybrid request mode as shown in Fig. 15. Additionally, reservation requests reduce the dependence of system performance on predicted requests. With the increase in the reservation rate, the total revenue, completion rate of requests, and unoccupied distance rate in the three predicted requests gradually approach and converge on the same point at the end.

In terms of reservation rate, it can be observed that when the request model changes from the single request mode (reservation rate is 0%) to the hybrid request mode, the proposed dispatching strategy continues to perform well with the total revenue and completion rate of requests remaining at a high level. It is remarkable that the operational performance is not good when the reservation rate is 100%. According to our predicted request generation rule, when the reservation rate is 100%, the immediate requests account for 0%, and it is unnecessary to generate predicted requests. Based on our statistics, there are 19 695 short-term reservation requests that account for 40.5% of the total generated requests. The large number of short-term reservation requests, whose request times are close to their departure time window, also require some predictions in advance to rebalance the supply and demand of vehicles. In practical application, it should be attempted to obtain not only the prediction of immediate requests but also that of short-term requests to improve operational performance.

6.5. Experiment 4

Experiment 4 tests our dispatching strategy in hybrid request mode where the reservation rate is 40% in the three fleet sizes (300/500/700) and analyzes the system sensitivity on the average advance reservation time.

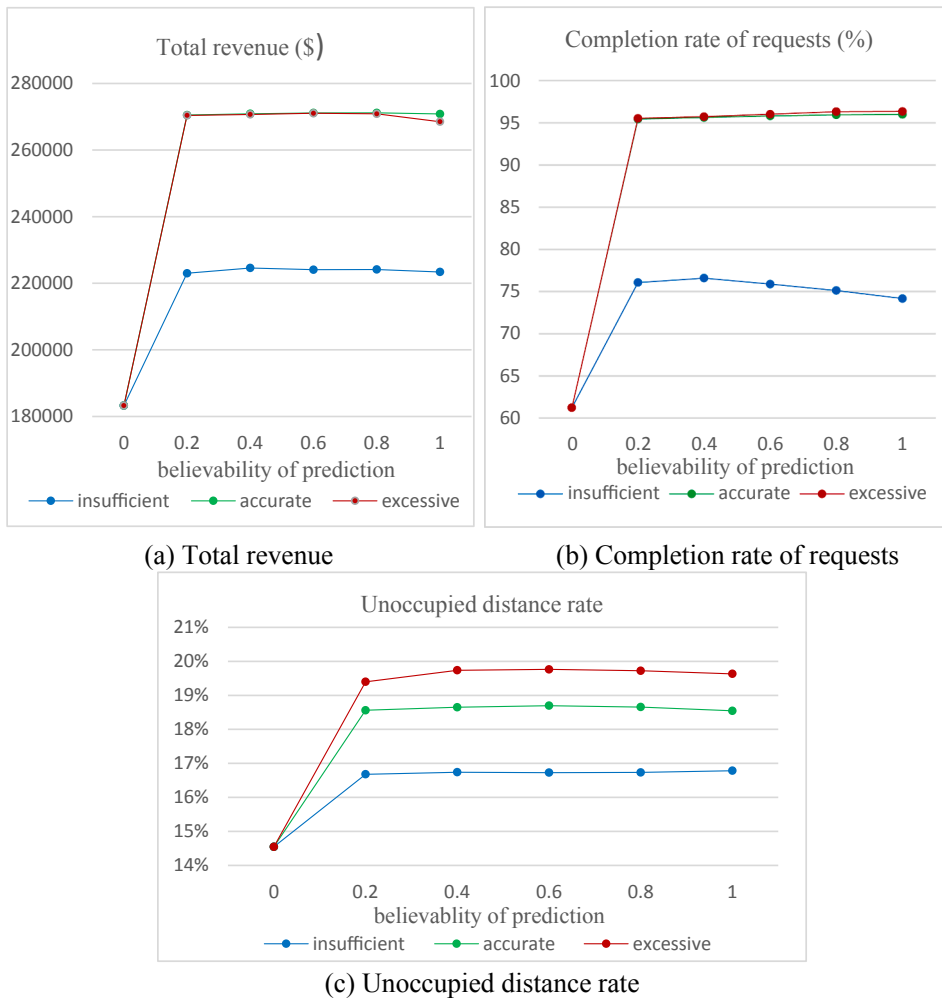


Fig. 14. Operational performance in three predicted requests in single request mode.

From the experimental results shown in Figs. 16–19, it is observed that aTaxi with a larger fleet size can achieve higher total revenues and completion of requests with lower unoccupied distance rates. In particular, when the fleet size is 700, practically all travelers can be served with the unoccupied distance rate remaining at a low level. An aTaxi can deliver 69 travelers on average.

In terms of the sensitivity on average advance time of reservation, the total revenue and completion of requests have been improved with the average advance time of reservation increasing when the fleet size is adequate, as shown in Figs. 16 and 17. Obtaining the detailed information of requests earlier is beneficial for the dispatcher to make a more reasonable decision, which results in better performance.

When the fleet size is inadequate, however, the total revenue and completion of requests present downward trends, as shown in Fig. 18, and the unoccupied distance rate increases with increasing average advance time, as shown in Fig. 19. This result exposes the disadvantage of our assignment policy of long-term reservation requests when the number of vehicles is insufficient. When a long-term reservation is requested, it will be accepted provided there is a vehicle that can reach its origin within the departure time window, and the fleet thereafter has a reserved capacity for service until the traveler is picked up. Its profitability and the space–time occupancy cost of the vehicle are not considered; however, they are extremely important particularly when the fleet size is inadequate. For example, a traveler reserves an aTaxi service from a remote location with a departure time that is distant from the present. The advance time of reservation is overly long that there must be a vehicle that can reach the remote pickup location place within the departure time window of the request. The request is therefore accepted according to our assignment policy. The acceptance of this request, however, is unreasonable because the space–time occupancy of the vehicle is costly considering the long pickup and long relocation distances to complete the request. If this request is rejected, the fleet may serve more requests and acquire more revenue. In contrast, the assignment policy of short-term requests is better. A short-term request will be accepted under two conditions: the fleet has the capacity to serve it, and it can realize maximum revenue in the short-term horizon as the objective

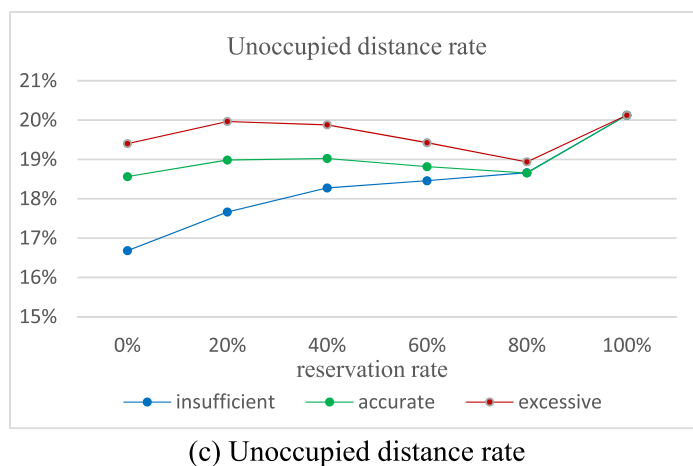
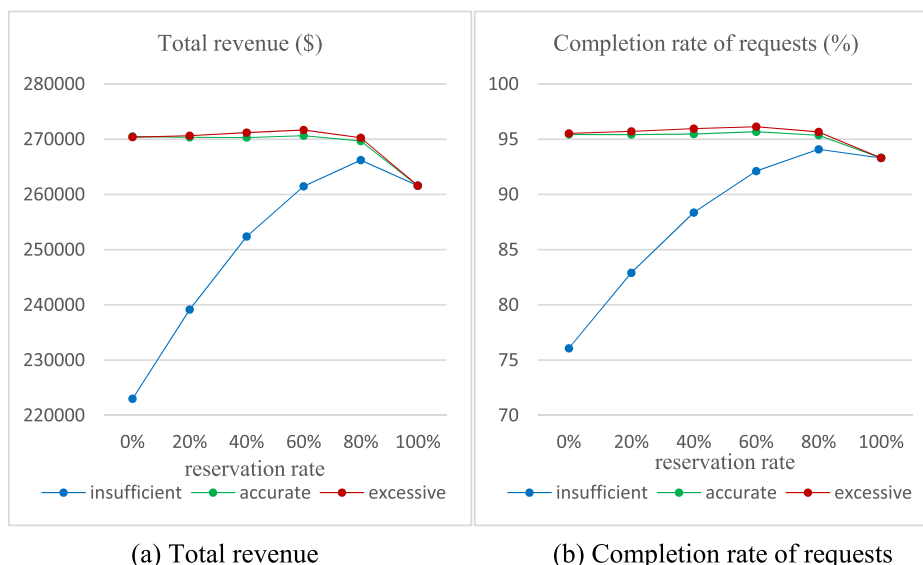


Fig. 15. Operational performance in three predicted requests in hybrid request mode.

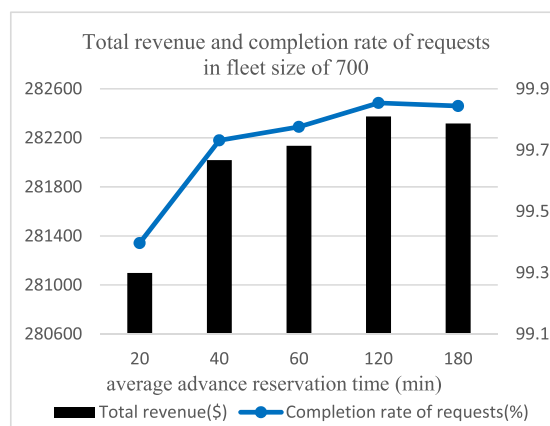


Fig. 16. Total revenue and completion rate of requests in fleet size of 700.

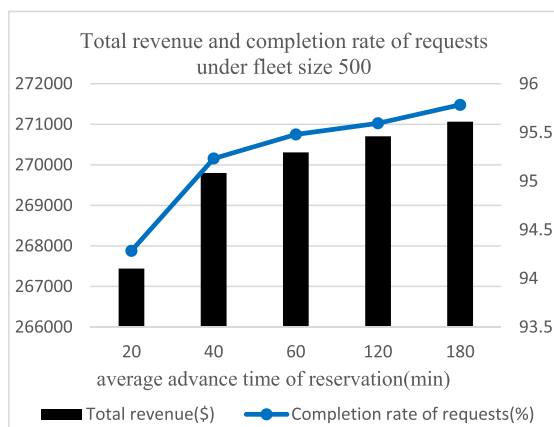


Fig. 17. Total revenue and completion rate of requests in fleet size of 500.

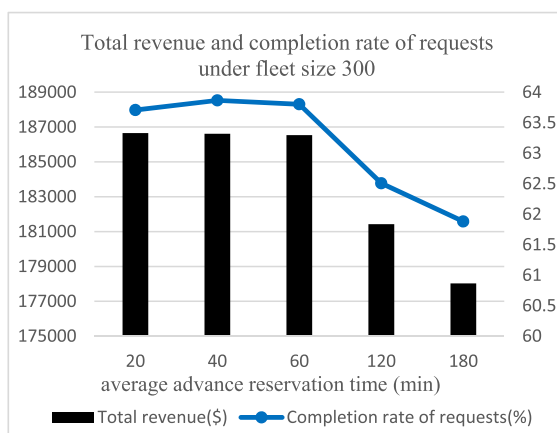


Fig. 18. Total revenue and completion rate of requests in fleet size of 300.

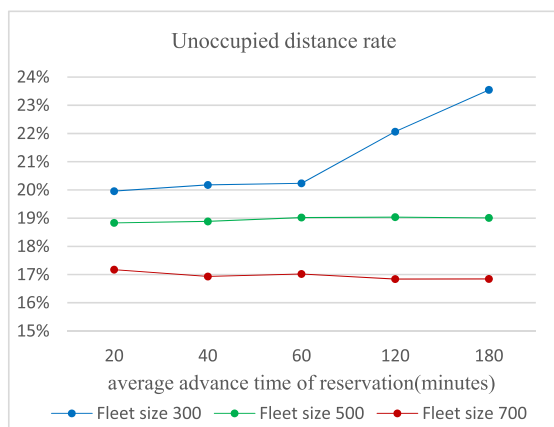


Fig. 19. Unoccupied distance rate in three fleet sizes.

function given by Eq. (4). Under this assignment policy, some unprofitable short-term requests will be rejected to reserve the capacity for other more profitable short-term requests. The longer advance time of reservation, which means more reservation requests are transformed from short-term requests to long-term requests, therefore results in a worse performance, as shown in Figs. 18 and 19. Considering the difficulty of evaluating the space–time occupancy cost of vehicles at different times, the assignment policy of long-term requests will be improved in future research.

7. Conclusions and future research

Autonomous taxis present a prominent advantage in empty vehicle rebalance and operation in hybrid request mode, having a considerable potential in improving service quality and economic efficiency. In this study, two aTaxi dispatching strategies are proposed, one for the single request mode and another for the hybrid request mode. In single request mode, the proposed dispatching strategy builds an aTaxi-request network based on the aTaxi information and requests that are acquired in real time. A network flow model is proposed for the aTaxi-request network to centrally plan the route for each aTaxi, including the route for empty vehicle rebalance. In hybrid request mode, a centralized dispatcher and decentralized autonomous dispatchers are designed to plan short-term and long-term routes for aTaxis, respectively. The centralized dispatcher extends the aTaxi dispatching strategy to the single request mode with the assignment of short-term reservation requests. The decentralized autonomous dispatcher is responsible for the assignment of long-term reservation requests. It also checks and adjusts the route to keep it feasible after central planning is conducted by the centralized dispatcher. This dispatching strategy guarantees that newly arrived requests can be responded to immediately, and all accepted requests will be served as travelers expect regardless of how the planned route changes.

In addition, the dispatching strategies are tested in a set of experiments. The results show that the proposed dispatching strategy in single request mode is superior to other myopic dispatching strategies in terms of service quality and economic efficiency. In hybrid request mode, the proposed dispatching strategy can maintain the high level of total revenue and completion rate of requests. Moreover, the increase in reservation rate can enhance the robustness of our method with respect to errors in predicted requests. A longer advance time of reservation requests can improve the revenue and completion rate of requests when the number of vehicles is adequate.

This study, however, has certain limitations. The dispatching strategy requires the estimated travel time between two points, which is acquired by static speed in this paper. But the travel time is dynamic and uncertain in real life due to complex traffic environment. Therefore, the dispatching strategy may not ensure the accepted request to be picked up within its departure time window. This problem will be studied in the future. The maintenance and charge or refueling behavior of the aTaxi are ignored. The assignment policy of long-term reservation has to be improved by considering the space–time occupancy cost of vehicle. The runtime of the centralized dispatcher is slightly longer in real-time vehicle dispatching, although the decentralized autonomous dispatchers improve computational efficiency. In future research, a more efficient algorithm will be designed to solve our model instead of using CPLEX, and we will attempt to conduct large-scale experiments. In view of the benefit of ridesharing, which has also been confirmed by numerous studies published in literature (Agatz et al., 2012, Furuhashi et al., 2013, Santi et al., 2014), extending our dispatching strategy to ride sharing will also be a part of our future work.

CRedit authorship contribution statement

Leyi Duan: Methodology, Software, Investigation, Data curation, Writing - original draft. **Yuguang Wei:** Conceptualization, Supervision, Project administration, Funding acquisition. **Jinchuan Zhang:** Validation, Writing - review & editing. **Yang Xia:** Formal analysis, Visualization.

Declaration of Competing Interest

None.

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